

Problem 16.13

A worker aged 40 wishes to accumulate a fund for retirement by depositing \$1,000 at the beginning of each year for 25 years. Starting at age 65 the worker plans to make 15 annual withdrawals at the beginning of each year. Assuming all payments are certain to be made, find the amount of each withdrawal starting at age 65 to the nearest dollar, if the effective rate of interest is 8% during the first 25 years but only 7% thereafter.

Solution

The worker deposits \$1,000 at the beginning of each year for 25 years, so the accumulated value at age 65 is

$$1000\ddot{s}_{\overline{25}|}$$

at 8% effective interest.

Let X be the annual withdrawal. Since withdrawals are made at the beginning of each year for 15 years, the value needed at age 65 is

$$X\ddot{a}_{\overline{15}|}$$

at 7% effective interest.

Equating the fund value at age 65 gives

$$1000\ddot{s}_{\overline{25}|} = X\ddot{a}_{\overline{15}|}.$$

Solving for X ,

$$X = \frac{1000\ddot{s}_{\overline{25}|}}{\ddot{a}_{\overline{15}|}}.$$

Using

$$\ddot{s}_{\overline{n}|} = (1+i)s_{\overline{n}|} \quad \text{and} \quad \ddot{a}_{\overline{n}|} = (1+i)a_{\overline{n}|},$$

we get

$$\begin{aligned} X &= \frac{1000(1.08) \left(\frac{(1.08)^{25} - 1}{0.08} \right)}{(1.07) \left(\frac{1 - (1.07)^{-15}}{0.07} \right)} \\ &\approx 8101.6546. \end{aligned}$$

Hence, to the nearest dollar, the amount of each withdrawal is

$$\boxed{\$8,102}.$$